

NAGOYA AP, Japan

WMO: 476350

Lat: 35.255N

Lon: 136.924E

Elev: 16

StdP: 101.14

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -1.0	(d) -10.1	(e) 1.6	(f) 5.1	(g) -8.9	(h) 1.8	(i) 4.8	(j) 10.9	(k) 7.0	(l) 10.1	(m) 6.7	(n) 2.0	(o) 340	(p) 0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.0	(c) 35.2	(d) 25.3	(e) 34.1	(f) 25.0	(g) 32.9	(h) 24.6	(i) 26.6	(j) 31.9	(k) 26.1	(l) 31.1	(m) 25.6	(n) 30.5	(o) 4.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	28.7	24.9	20.0	28.6	24.1	19.1	28.1	83.3	32.2	81.0	31.1	78.9	30.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	8.6	7.5	-4.2	37.4	1.3	1.2	-5.1	38.3	-5.8	39.0	-6.5	39.7	-7.5	40.6
			-5.4	27.5	1.0	0.7	-6.1	28.0	-6.6	28.4	-7.2	28.8	-7.9	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	4.5	5.5	9.1	14.7	19.8	23.6	27.8	29.1	25.1	19.1	12.7	7.1	
	DBStd	8.84	2.20	2.77	3.08	3.23	2.43	2.30	2.49	2.08	2.81	2.85	3.08	2.85	
	HDD10.0	463	170	128	54	4	0	0	0	0	0	0	9	97	
	HDD18.3	1742	428	358	286	114	13	0	0	0	1	24	171	347	
	CDD10.0	2863	0	3	28	144	305	408	551	592	453	283	89	8	
	CDD18.3	1100	0	0	0	5	59	158	293	333	203	48	1	0	
	CDH23.3	10455	0	0	1	34	353	1032	3164	3972	1688	209	1	0	
CDH26.7	4128	0	0	0	3	67	282	1351	1852	549	24	0	0		
Wind	WSAvg	3.1	3.2	3.5	3.7	3.5	3.3	2.8	3.0	3.0	2.9	2.8	2.7	2.9	
Precipitation	PrecAvg	1560	47	64	112	138	155	199	207	146	224	133	78	48	
	PrecMax	2033	155	158	211	230	331	514	450	379	708	530	230	125	
	PrecMin	905	1	9	3	7	51	58	29	4	31	10	7	0	
	PrecStd	258	33	37	47	51	63	80	105	95	129	92	48	30	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.0	21.0	26.2	31.0	33.2	37.0	37.2	34.1	29.0	22.9	17.8	
		MCWB	7.6	11.1	12.0	15.2	18.3	23.3	25.6	25.6	24.8	20.9	16.0	12.7	
	2%	DB	11.9	14.1	18.8	24.1	28.2	31.1	35.2	35.9	32.8	27.1	21.1	15.1	
		MCWB	6.8	8.3	11.3	15.2	17.6	21.9	25.2	25.4	23.9	19.8	14.5	10.1	
	5%	DB	10.2	12.2	16.9	22.2	26.9	29.8	34.0	34.8	31.1	25.8	19.8	13.8	
		MCWB	5.8	7.2	10.8	14.2	17.4	21.4	25.1	25.3	23.3	19.0	14.1	9.3	
	10%	DB	9.1	11.0	15.1	20.8	25.1	28.1	32.8	33.2	29.9	24.1	18.1	12.1	
		MCWB	4.9	6.3	9.3	14.0	16.9	20.6	24.8	24.9	22.7	18.0	13.1	8.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.2	15.3	19.9	22.2	25.4	27.2	27.5	26.4	23.7	17.9	13.9
			MCDB	11.3	15.8	17.1	21.5	25.1	29.3	33.2	33.4	30.7	25.7	20.1	15.4
2%		WB	7.7	10.1	13.5	17.7	20.6	24.5	26.5	26.8	25.5	22.0	16.5	11.5	
		MCDB	9.9	12.7	16.6	20.7	23.9	28.2	32.0	32.1	29.2	25.2	19.2	14.0	
5%		WB	6.5	8.3	12.0	16.4	19.4	23.6	25.9	26.4	24.8	20.5	15.4	9.9	
		MCDB	9.1	11.0	15.7	20.3	23.5	27.1	31.1	31.5	28.7	23.7	18.2	12.6	
10%		WB	5.4	6.7	10.3	15.2	18.4	22.4	25.4	25.9	24.2	19.3	14.2	8.6	
		MCDB	8.4	10.1	14.0	18.9	23.0	26.2	30.4	30.9	28.1	22.8	17.1	11.6	
Mean Daily Temperature Range		5% DB	MDBR	8.2	8.9	9.5	9.9	9.4	7.7	7.5	8.0	7.7	8.4	8.7	8.2
			MCDBR	9.8	10.8	11.9	12.2	12.3	10.0	9.5	9.5	8.9	9.5	10.0	9.7
	MCWBR		6.3	6.8	6.9	5.9	5.0	3.6	3.1	2.9	3.2	4.4	5.6	6.1	
	5% WB	MCDBR	8.4	8.8	9.4	8.6	8.4	7.1	8.1	8.1	7.1	7.2	7.6	8.0	
		MCWBR	6.2	6.7	7.3	5.7	5.1	3.4	3.1	3.1	3.2	4.5	5.4	6.1	
Clear-Sky Solar Irradiance	taub	0.391	0.430	0.497	0.540	0.549	0.578	0.567	0.545	0.496	0.458	0.415	0.390		
	taud	2.163	2.070	1.920	1.867	1.914	1.913	1.996	2.045	2.149	2.173	2.200	2.189		
	Ebn at Noon	781	793	770	760	760	736	741	749	766	758	750	754		
	Edh at Noon	116	143	181	200	193	193	177	165	142	128	112	107		
All-Sky Solar Radiation	RadAvg	2.33	3.04	3.99	4.77	5.24	4.73	4.88	5.02	4.01	3.27	2.51	2.07		
	RadStd	0.16	0.23	0.28	0.45	0.57	0.42	0.69	0.47	0.36	0.32	0.21	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) +0.32	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +107	(m) +69
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page